Industrial Internship Report on"BankingInformationSystem**"**

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was "Banking Information System", a console-based application developed using Core Java to simulate the basic operations of a banking system. The project enables users to create bank accounts, deposit and withdraw money, check account balances, and view account details through a user-friendly menu-driven interface. It was designed by applying Object-Oriented Programming (OOP) concepts such as classes, objects, encapsulation, and inheritance, along with exception handling and modular programming. The project provided practical experience in software development, problem-solving, debugging, and implementing real-world banking functionalities while strengthening my understanding of Core Java programming. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

During the six-week industrial internship, I worked on the **Banking Information System** project using **Core Java**. The internship was structured to provide practical exposure to software development by designing and implementing a real-world application.

In the initial weeks, I studied the project requirements, analyzed the problem statement, and designed the system architecture. I identified the key functionalities of a banking system and planned the application using Object-Oriented Programming (OOP) principles.

In the development phase, I implemented core modules such as account creation, deposit, withdrawal, balance inquiry, account details, and transaction management. The application was developed as a menu-driven console program with proper input validation, exception handling, and modular code structure to ensure reliability and maintainability.

During the testing phase, I verified the functionality of each module using different test cases, corrected errors through debugging, and improved the overall performance and user experience of the application. The project code was organized using GitHub for version control and documentation.

Throughout the internship, I strengthened my understanding of **Core Java, Object-Oriented Programming, exception handling, debugging, modular programming, and software development practices**. This project enhanced my problem-solving abilities and provided valuable hands-on experience in developing a practical software solution that simulates basic banking operations. Overall, the internship was a rewarding learning experience that improved both my technical knowledge and professional skills, preparing me for future software development projects and industry challenges.

A relevant industrial internship plays a vital role in career development by bridging the gap between academic learning and real-world industry practices. It enables students to apply theoretical concepts to practical projects, enhancing their technical knowledge and problem-solving abilities.

Through this internship, I gained hands-on experience in developing a **Banking Information System** using Core Java, which strengthened my understanding of programming concepts, software design, debugging, and project development. Working on an industry-oriented project also improved my analytical thinking, time management, communication, and teamwork skills.

Additionally, the internship familiarized me with professional development practices such as project planning, testing, documentation, and version control using GitHub. These experiences have increased my confidence, improved my employability, and prepared me to meet the expectations of the software industry. Overall, this internship has been an important step in building the technical and professional skills required for a successful career in information technology.

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The project undertaken during the internship was **"Banking Information System"**, developed using **Core Java**. The objective of the project was to design and implement a console-based application that automates basic banking operations, replacing the traditional manual approach with a secure and efficient digital system.

The problem addressed by this project is the inefficiency and possibility of errors associated with maintaining banking records manually. Manual processes are time-consuming, prone to calculation mistakes, and make retrieving customer information difficult. The proposed system provides a simple, reliable, and user-friendly solution for managing banking activities.

The application allows users to perform essential banking operations such as **creating customer accounts, depositing and withdrawing money, checking account balances, viewing account details, and managing transactions** through a menu-driven interface. The project was developed using **Object-Oriented Programming (OOP)** concepts, modular programming, and exception handling to ensure code reusability, maintainability, and robustness.

This project provided practical exposure to software development life cycle (SDLC), Core Java programming, debugging, testing, and documentation, while demonstrating how programming can be used to solve real-world business problems effectively.

1.1.1 Opportunity Given by USC/UCT

The internship organized by **upskill Campus (USC)** in collaboration with **UniConverge Technologies Pvt. Ltd. (UCT)** provided an excellent opportunity to work on an industry-oriented project. The program enabled me to apply my academic knowledge to a real-world problem while enhancing my practical skills in software development. It also exposed me to professional project development practices, documentation, testing, and version control, preparing me for future industry challenges.

1.1.2 How the Program Was Planned

The internship was conducted over **six weeks** in a structured manner. The initial phase focused on understanding the project requirements and planning the solution. The development phase involved implementing the Banking Information System using Core Java, followed by testing, debugging, and improving the application's functionality. The final phase included project documentation, performance evaluation, and submission of the completed project and report.

1.1.3 My Learnings and Overall Experience

This internship significantly enhanced my technical knowledge and practical skills. I gained hands-on experience in Core Java, Object-Oriented Programming, exception handling, debugging, and software testing. I also learned the importance of project planning, code organization, documentation, and GitHub for version control. Overall, the internship was a valuable learning experience that strengthened my confidence and prepared me for future software development roles.

1.1.4 Acknowledgement

I sincerely express my gratitude to **upskill Campus (USC)** and **UniConverge Technologies Pvt. Ltd. (UCT)** for providing this valuable internship opportunity. I am thankful to all the mentors, trainers, and coordinators who guided and supported me throughout the program. I also thank my college faculty members for their encouragement and my family and friends for their constant motivation and support during the successful completion of this project.

1.1.5 Message to My Juniors and Peers

I encourage my juniors and peers to actively participate in internships and practical projects, as they provide valuable industry exposure beyond classroom learning. Be curious, practice consistently, and never hesitate to learn from challenges and mistakes. Developing technical skills along with discipline, teamwork, and continuous learning will help build a strong foundation for a successful career in the IT industry.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

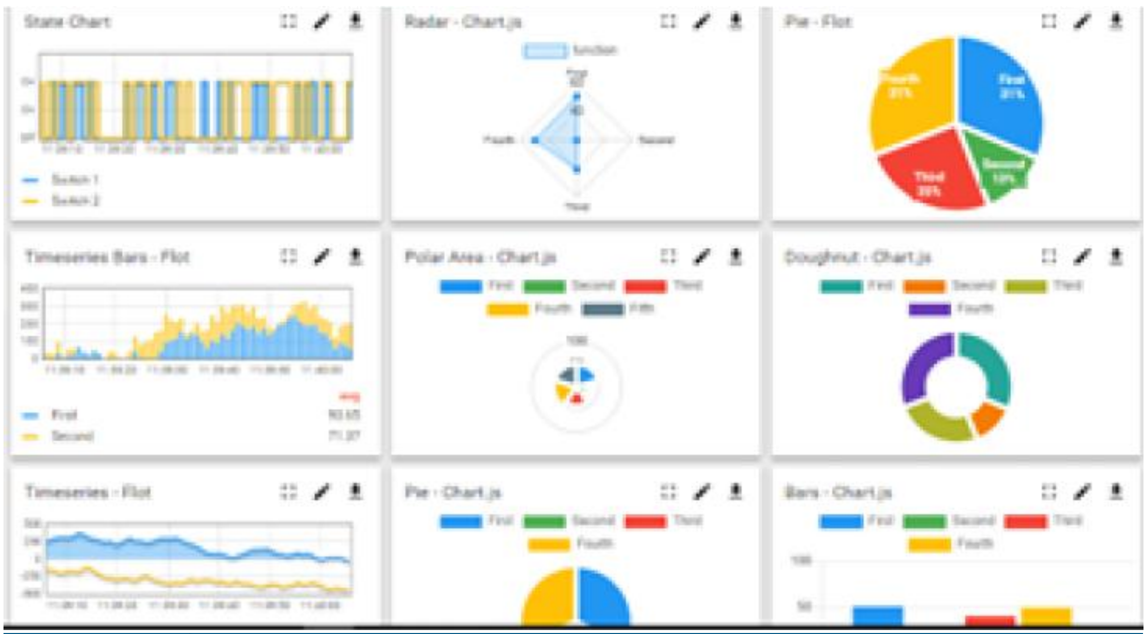
i. UCT IoT Platform (uct Insight)

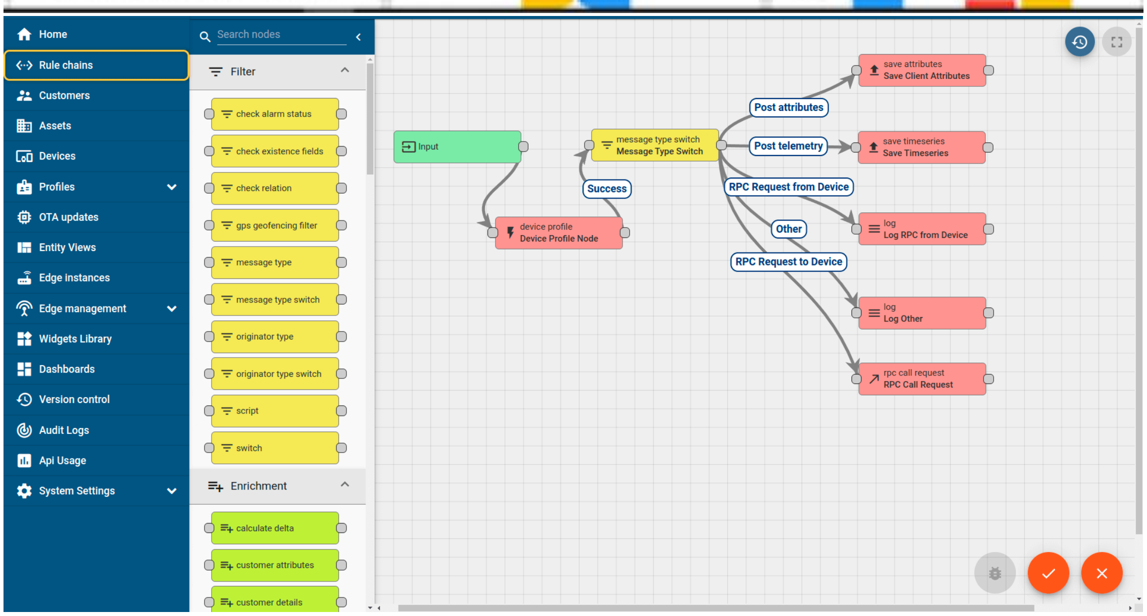
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

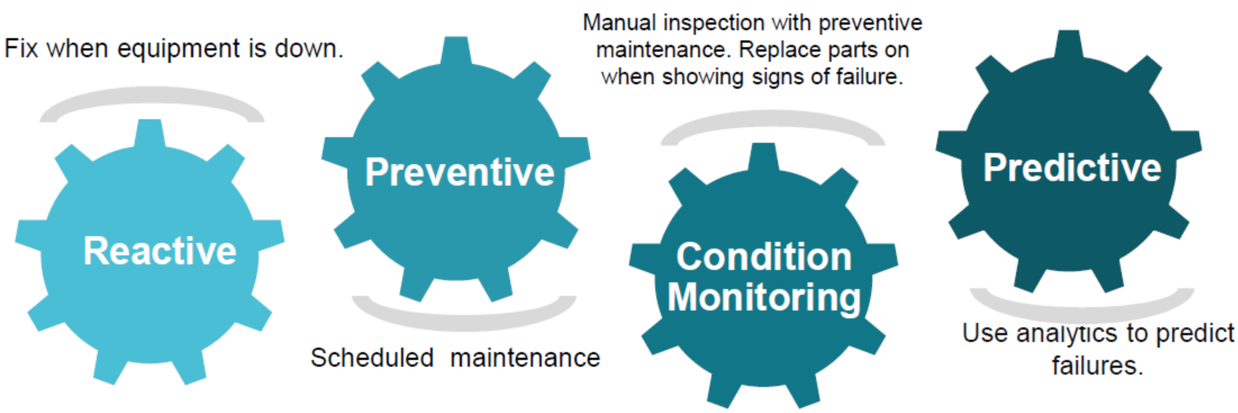
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

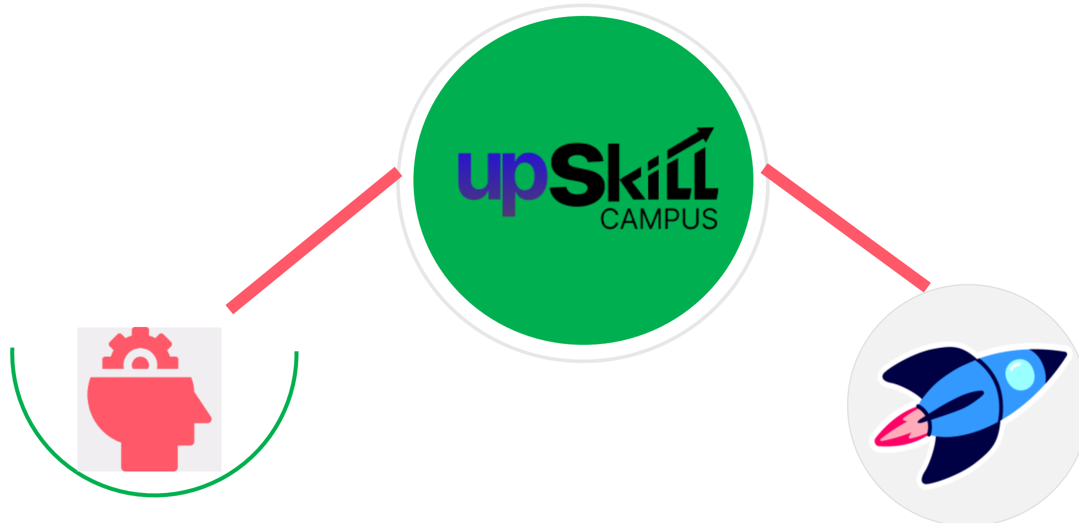
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

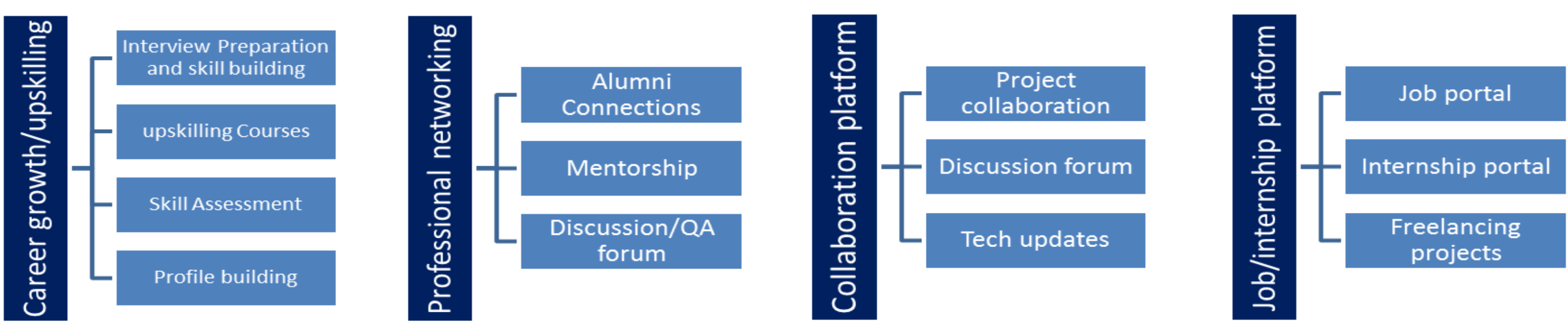
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

Problem Statement

The assigned problem statement was to develop a **Banking Information System** using **Core Java** that demonstrates the fundamental operations of a real banking application. The objective was to create a menu-driven console application capable of performing essential banking services in an efficient and user-friendly manner.

Traditional manual banking record management is often time-consuming, prone to human errors, and difficult to maintain as the volume of customer data increases. Retrieving account information, updating balances, and maintaining transaction records manually can lead to inaccuracies and reduced efficiency.

To address these challenges, the proposed solution automates basic banking operations such as **creating new customer accounts, depositing and withdrawing funds, checking account balances, viewing account details, and maintaining transaction records**. The application was developed using **Object-Oriented Programming (OOP)** principles, modular programming, and exception handling to ensure reliability, code reusability, and ease of maintenance.

The project provides a simple prototype of a banking system that demonstrates how Core Java can be used to develop practical, secure, and efficient software solutions for real-world business applications while enhancing programming, debugging, and software development skills.

3 Existing and Proposed solution

3.1.1 4.1 Existing Solutions

Several banking management applications and manual record-keeping systems are currently used to manage customer accounts and transactions. While these systems perform basic banking operations, they have certain limitations, especially in small-scale organizations and educational environments.

Limitations of Existing Solutions:

- Manual record management is time-consuming and prone to human errors.
- Commercial banking software is expensive and often includes complex features that are unnecessary for learning purposes.
- Many existing systems require database servers and internet connectivity, making them less suitable for beginners.
- Updating and maintaining customer records manually can lead to inconsistencies and data redundancy.
- Existing solutions provide limited opportunities for students to understand the implementation of banking operations using programming concepts.

3.1.2 4.2 Proposed Solution

The proposed solution is a **Core Java-based Banking Information System** that provides a simple, menu-driven interface for managing basic banking operations. The application demonstrates the practical implementation of Object-Oriented Programming (OOP) concepts while simulating real-world banking functionalities.

The system includes the following features:

- Customer account creation
- Deposit and withdrawal of money
- Balance inquiry
- Viewing account details
- Transaction management
- Input validation and exception handling
- Modular and reusable code structure

The application is lightweight, easy to use, and can be executed without requiring external databases or internet connectivity, making it suitable for educational purposes and software development practice.

3.1.3 4.3 Value Addition

The proposed system provides several improvements over traditional approaches:

- Reduces manual effort and minimizes calculation errors.
- Demonstrates real-world application of Core Java and OOP concepts.
- Provides a user-friendly menu-driven interface for easy operation.
- Implements exception handling for reliable and secure execution.
- Uses modular programming, making the code easier to maintain and extend.
- Can be upgraded in the future by integrating a database (JDBC/MySQL), graphical user interface (JavaFX/Swing), user authentication, and online banking features.

This project serves as a strong foundation for understanding banking software development while improving programming, debugging, and software engineering skills.

3.2 Code submission https://github.com/Princekr001

3.3 Report submission https://github.com/Princekr001/upskillcampus

4 Proposed Design/ Model

The **Banking Information System** was designed using a modular and object-oriented approach in **Core Java**. The system follows a simple workflow where the user interacts with a menu-driven interface to perform banking operations. Each request is processed by the appropriate module, validated, and executed before displaying the result to the user.

The design begins with the **Main Menu**, where users can select operations such as account creation, deposit, withdrawal, balance inquiry, account details, and exit. Based on the selected option, the request is forwarded to the **Bank Service** module, which contains the business logic for processing banking operations.

The Bank Service interacts with the **User (Account) Model**, where customer information such as account number, account holder name, balance, and transaction details are stored. Before performing any operation, the system validates the user input and account information. If the validation is successful, the requested operation is executed and the updated information is displayed. Invalid inputs or exceptions are handled gracefully using Java exception handling mechanisms.

The overall design follows the principles of **Object-Oriented Programming (OOP)**, including encapsulation, modularity, and code reusability. This architecture makes the application easy to understand, maintain, and extend with future enhancements such as database integration, graphical user interface, authentication, and online banking features.

4.1 High Level Diagram (if applicable)

```
+--------------------+
|       User         |
+---------+----------+
          |
          v
+--------------------+
|  Main Menu (UI)    |
+---------+----------+
          |
          v
```

```
+--------------------+
|   Bank Service     |
| (Business Logic)   |
+---------+----------+
          |
          v
+--------------------+
| User/Account Model |
+---------+----------+
          |
          v
+--------------------+
| Display Result     |
+--------------------+
```

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

4.2 Low Level Diagram (if applicable)

```
Start
  |
Display Main Menu
  |
```

```
Select Operation
       |
+------------------------------+
| Create Account               |
| Deposit Money                |
| Withdraw Money               |
| Check Balance                |
| View Account Details         |
| Exit                         |
+------------------------------+
       |
Validate Input
       |
Perform Requested Operation
       |
Update Account Details
       |
Display Output
       |
Exit / Return to Menu
```

4.3 Interfaces (if applicable)

The Banking Information System provides a **console-based, menu-driven interface** that allows users to perform banking operations efficiently. The application interacts with the user through keyboard input and displays the results on the console.

4.4 User Interface

```
=========================================
   BANKING INFORMATION SYSTEM
=========================================

1. Create Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. View Account Details
6. Exit

Enter your choice:.
```

4.4 Block Diagram

```
+----------------+
|      User      |
+-------+--------+
        |
```

```
          v
+---------------------+
|  Console Interface  |
+----------+----------+
           |
           v
+---------------------+
|   Banking Service   |
|  (Business Logic)   |
+----------+----------+
           |
           v
+---------------------+
|  Account Database   |
| (Java Objects/List) |
+----------+----------+
           |
           v
+---------------------+
| Display Output      |
+---------------------+
```

5 Performance Test

The **Banking Information System** was tested to evaluate its correctness, reliability, and performance under different user inputs. Although the project is a console-based prototype developed using Core Java, it was designed by considering practical software engineering principles such as input validation, modularity, and error handling. The performance testing ensured that all banking operations executed correctly and efficiently.

5.1 Test Plan/ Test Cases

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
TC-01	Create a new account	Account created successfully	Account created successfully	Pass
TC-02	Deposit valid amount	Balance updated correctly	Balance updated correctly	Pass
TC-03	Withdraw amount within balance	Amount deducted successfully	Amount deducted successfully	Pass
TC-04	Withdraw amount greater than balance	Display "Insufficient Balance"	Correct error message displayed	Pass
TC-05	Check account balance	Current balance displayed	Balance displayed correctly	Pass
TC-06	View account details	Customer information displayed	Information displayed correctly	Pass
TC-07	Enter invalid menu option	Display error message	Error handled successfully	Pass
TC-08	Enter invalid amount (negative value)	Reject input	Invalid input handled correctly	Pass

5.2 Test Procedure

- The application was compiled and executed using the Java Development Kit (JDK).
- Multiple test accounts were created with different account details.
- Each banking operation, including account creation, deposit, withdrawal, balance inquiry, and account viewing, was tested individually.
- Invalid inputs such as incorrect menu options, negative transaction amounts, and insufficient account balance were entered to verify exception handling.
- The output generated by the application was compared with the expected results to ensure correctness and reliability.
- After successful verification, the application was reviewed for code consistency and overall functionality.

5.3 Performance Outcome

The Banking Information System successfully executed all core banking operations with accurate results. The menu-driven interface responded quickly, and all test cases passed without major functional issues. Exception handling prevented the application from crashing due to invalid inputs, ensuring reliable execution.

Although the application stores data temporarily in memory, it performs efficiently for educational and small-scale use. The modular architecture makes the system easy to maintain and extend. Future enhancements such as **database integration (MySQL), graphical user interface (JavaFX/Swing), user authentication, and transaction logging** can improve scalability, security, and long-term usability for real-world applications.

Overall Result: The project met its functional objectives and demonstrated the practical implementation of Core Java concepts while providing a reliable prototype of a basic Banking Information System.

6 My learnings

The six-week industrial internship provided me with valuable practical experience and helped bridge the gap between academic learning and real-world software development. Working on the **Banking Information System** enabled me to apply Core Java concepts to develop a functional application while following a structured software development approach.

During this internship, I strengthened my understanding of **Object-Oriented Programming (OOP)**, including classes, objects, encapsulation, abstraction, inheritance, and polymorphism. I also improved my knowledge of **exception handling**, modular programming, input validation, and debugging techniques, which are essential for developing reliable and maintainable software.

The project enhanced my analytical thinking and problem-solving abilities by allowing me to design logical solutions for real-world banking operations. I also gained practical experience in **testing software, identifying and fixing errors, preparing technical documentation, and using GitHub for project management and version control**.

In addition to technical skills, the internship helped me develop important professional qualities such as **time management, planning, attention to detail, and continuous learning**. Completing the project within the given timeline improved my confidence in handling software development tasks independently.

Overall, this internship has strengthened my foundation in software development and prepared me for future opportunities in the IT industry. The knowledge and experience gained through this project will support my career growth by enabling me to build more advanced applications, contribute effectively to software development teams, and continuously adapt to new technologies and industry requirements.

7 Future work scope

The current **Banking Information System** successfully demonstrates the core functionalities of a banking application using Core Java. However, due to the limited duration of the internship, several advanced features could not be implemented. These enhancements can be incorporated in future versions to improve the system's functionality, scalability, and usability.

The following improvements are proposed for future development:

- **Database Integration:** Replace the in-memory data storage with **MySQL** using **JDBC** to provide permanent storage of customer and transaction records.
- **Graphical User Interface (GUI):** Develop an interactive interface using **JavaFX** or **Swing** to improve the user experience.
- **User Authentication:** Implement secure login functionality with usernames, passwords, and role-based access control for administrators and customers.

- **Transaction History:** Maintain detailed transaction logs with date, time, transaction type, and account information for better record management.
- **Interest and Loan Management:** Add modules for savings accounts, fixed deposits, loan processing, and automatic interest calculation.
- **Online Banking Features:** Extend the application to support fund transfers, online payments, and account management through web or mobile platforms.
- **Security Enhancements:** Incorporate password encryption, secure authentication mechanisms, and audit logs to improve data security.
- **Notification System:** Integrate email or SMS notifications for successful transactions, account updates, and security alerts.

These enhancements would transform the current prototype into a more comprehensive and industry-ready banking application, capable of supporting real-world banking requirements while providing greater security, scalability, and user convenience.